Inaugural International Orbitrap-IRMS Users Meeting

Goldschmidt 2025 Pre-Conference Science Workshop

Prague, Czech Republic 5–6 July 2025

Convenors: Amy Hofmann (California Institute of Technology), Issaku Kohl (University of Utah), Andrea Erhardt (University of Kentucky)

Saturday, 5 July 2025

08:30 – 08:45	Introduction / Welcome / Day 1 Schedule and Goals [Amy]
08:45 – 09:15	Technical aspects of Orbitrap-IRMS overview [Issaku]
09:15 – 10:15	End-to-end demonstration of Orbitrap-IRMS measurements [Nils]
10:15 – 10:30	BREAK

Session #1: Small Molecule Analyses

10:30 – 11:15	**Talk #1:** Nora Bernet, “Oxygen isotope analyses of phosphate and organophosphorus compounds by Orbitrap mass spectrometry”
11:15 – 12:00	**Talk #2:** Shohei Hattori, “Isotope analysis of methanesulfonate in ESI-Orbitrap-IRMS”
12:00 – 13:00	LUNCH (+ informal discussions!)
13:00 – 14:00	**Group Discussion:** topics raised in Talks 1 & 2 and over lunch
14:00 – 15:30	POSTER SESSION + coffee/tea BREAK
15:30 – 17:00	Wrap-up Day 1 Discussion + Overview of Day 2

Sunday, 6 July 2025

08:30 – 08:45	Announcements / Day 2 Schedule and Goals [Amy]

Session #2: Large Molecules + Compound- and Position-Specific Analyses

08:45 – 09:30	**Talk #3:** Aoife Canavan, "Isotope analysis of sulfamethoxazole with ESI-Orbitrap: Standard preparation and LC coupling"
09:30 – 10:15	**Talk #4:** Lubna Shawar, "Site-specific isotopic fingerprinting of steroidal compounds: Applications in geochemistry and forensics"
10:15 – 10:30	BREAK
10:30 – 11:30	**Group Discussion:** topics raised in Talks 3 & 4

Session #3: Standardization, Data Handling, and Advanced Methods

11:30 – 12:15	**Talk #5:** Gabriella Weiss, "Analysis of intramolecular carbon isotope distributions in alanine and steps toward standardization within the iso-Orbi community"
12:15 – 13:15	LUNCH (+ informal discussions!)
13:15 – 14:00	**Talk #6:** Nils Kuhlbusch, "From sample to data: Advanced strategies for accurate isotopocule ratio analysis of organic molecules in complex samples using ESI-Orbitrap MS and on-line HPLC"
14:00 – 15:00	**Group Discussion:** topics raised in Talks 5 & 6 as well as over lunch
15:00 – 16:00	POSTER SESSION with coffee/tea BREAK
16:00 – 17:00	Workshop synthesis and community plans going forward [Organizers]

LOGISTICS

Venue: Charles University, Faculty of Science Albertov 6, Prague 2

View on Google Maps

Badging: Please plan on arriving by 08:15 on Saturday to pick up your badge in the building lobby.

Sustenance: We will be providing coffee and tea during the two breaks and a bagged lunch each day. Vegetarian and vegan meals will be provided for those who requested them at the time of registration. Please double-check your registration to make sure everyone receives their meal of choice. Additional snacks will not be provided, so be sure to pack something if you'd like to munch on something during the breaks.

Electronics: WiFi information will be provided on-site. Please remember to bring the appropriate power adapter with you.

Posters: We will have poster stands available in the classroom. These stands can accommodate posters with the same dimensions as those requested for the Goldschmidt Conference itself: A0 (~85 cm wide by ~115 cm high).

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FINAL THOUGHTS

One of our major goals for this meeting is to establish a collegial but also self-assessing community of practice centered on the use of Orbitrap-IRMS to push the boundaries of isotope chemistry. We are all in that space now, and, as we disseminate our results and discoveries, this field will undoubtedly grow. This workshop is an opportunity for us to share and discuss our work and the scientific frontier we each envision.

As a community, we already have much to learn from one another—each group is currently feeling out different aspects of Orbitrap-IRMS technology as it can be applied to their specific scientific interests. One of our other major goals for this meeting is to begin converging on "best practices" for isotope ratio analyses using Orbitrap-IRMS. As a community, we are just starting to sort out things like instrumental mass fractionation, how we standardize analyses, what constitutes a "useable" measurement, and how the 'answers' to these things differ across different Orbitrap platforms, different analytes, and different types of measurements. By no means will we solve these issues in just two days, but we need to start having those conversations now—and we need to continue having them as the community grows.

To those ends, we also hope that this meeting with serve as a nexus for establishing new connections and potentially new collaborations across the Orbitrap-IRMS community. At present, I know of over 150 individuals actively engaged in Orbitrap-IRMS research, and I am quite sure there are many more—and many more interested in joining our field. We here are obviously only a small fraction of that burgeoning community. Thus, another of our major goals is to leave this meeting with plans in place for convening the next one…and the next one…and the next one. What our future meetings entail, when and where they occur, and who leads them are all things to be discussed this weekend.

We strongly encourage all attendees—regardless of career stage or experience with Orbitrap-IRMS—to engage in our group discussions or, if more comfortable, in one-on-one or small group conversations during breaks. Although we have no formal plans for a closing dinner, we also hope that your conversations continue beyond this weekend. As many of you know, I (Amy) maintain a list of Orbitrap-IRMS users and those interested in adopting the technology. I periodically use this list to disseminate relevant information (like announcing this workshop) via email. If you are not on that list but wish to be, please send me a note; likewise, feel free to pass that information along to other colleagues who might be interested.

On behalf of my co-convenors, I sincerely hope that you find the meeting intellectually stimulating as well as fun. It is a real privilege for me to be involved with a community of like-minded scientists in these early stages of development. The Orbitrap-IRMS community will be what we as a community shape it to be. What could be more thrilling than to be at the frontier?

Amy E. Hofmann

27 June 2025